

# AUXILIATURA DE ESTRUCTURAS HIPERESTÁTICA

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CIV 246 PARALELO 4D1

# ANALISIS ESTRUCTURAL

VIGA ISOSTATICA

$$V = 0$$

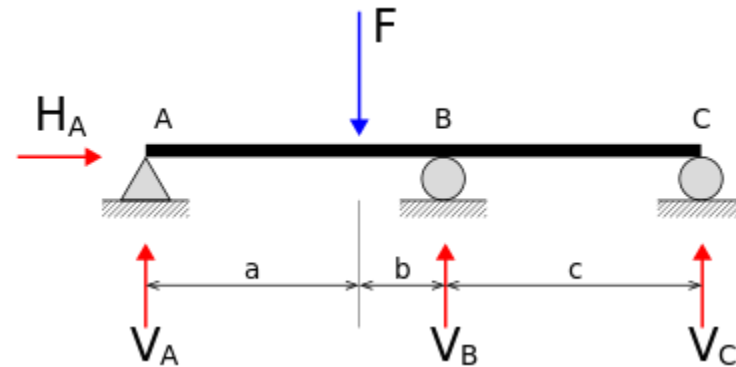


VIGA HIPOESTATICA

$$V < 0$$

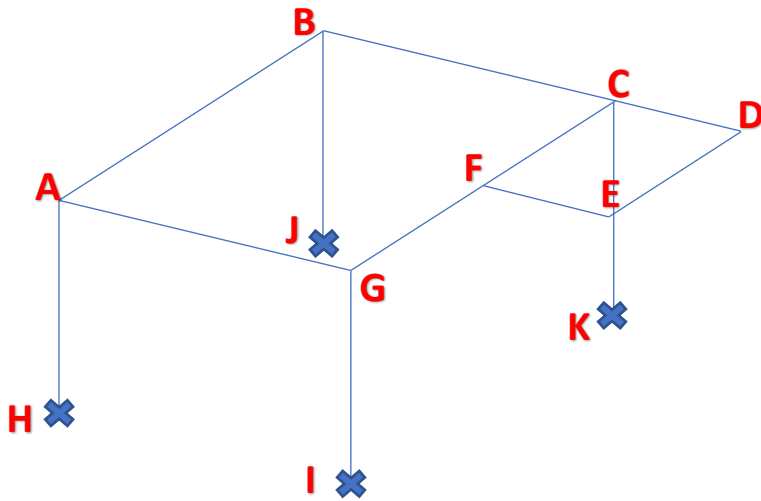
VIGA HIPERESTATICA

$$V > 0$$



# ESTRUCTURAS ESPACIALES

## ESTRUCTURA APORTICADA



$$GH = C - GL$$

$$GH = (CE + CI) - (GLI + GLE)$$

$$CE = 6 * N^{\circ} \text{APOYOS} = 6 * 4 = 24$$

$$\sum Ci = 6(b - 1) = 6(3 - 1)A, B, F, G(4) = 48$$

$$= 6(4 - 1)C(1) = 18$$

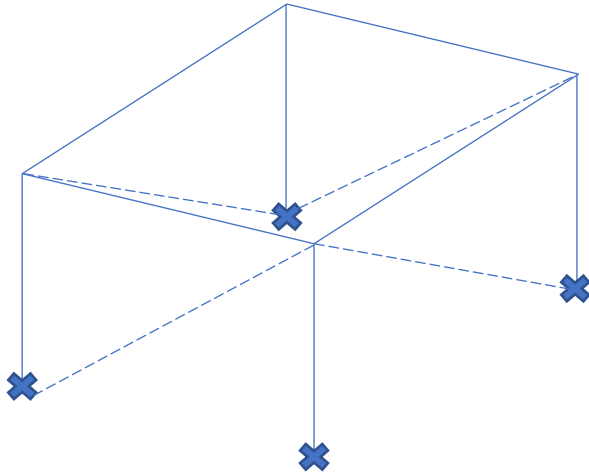
$$= 6(2 - 1)D, E(2) = 12$$

$$GLI = 6(C - 1) = 6(12 - 1) = 66$$

$$GLE = 6$$

$$GH = (24 + 78) - (66 + 6) = 30 \text{ ES HIPERESTATICO}$$

# ESTRUCTURAS RETICULARES



$$GH = GHI + GHE$$

$$GHI = b - (3n - 6) = 12 - (3 * 4 - 6) = 6$$

$$GHE = 3n_a - 6 = 3(4) - 6 = 6$$

$$GH = 6 + 6 = 12 \quad \text{HIPERESTATICA}$$

$n$  = NUMERO DE NUDOS ARTICULADOS

$b$  = NUMERO DE BARRAS

$GHI$  = GRADO DE HIPERESTATICIDAD INTERNO

$GHE$  = GRADO DE HIPERESTATICIDAD EXTERNO

# TRES MOMENTOS

1)APOYO SIMPLE  $M=0$

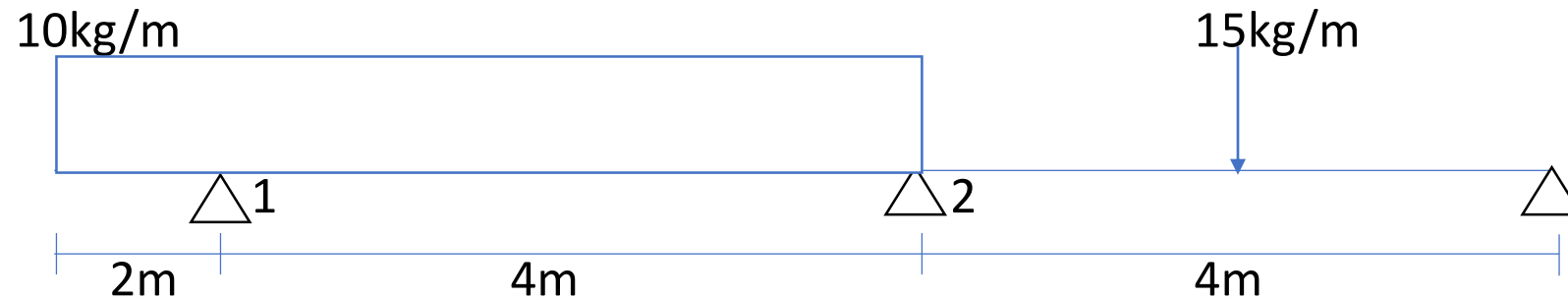
2)EMPOTRAMIENTO ALARGAMOS UN TRAMO VIRTUAL O IMAGINARIO DONDE  
TODOS LOS VALORES SON IGUALES A 0

3)VOLADIZO  $M=\text{FUERZA} \times \text{BRAZO}$

LAS TRES CONDICIONES SON PARA APOYOS EN EXTREMOS

AUXILIAR:CRISTIAN VELASQUEZ LOZANO

Ejemplo 2) determinar el diagrama de momentos y cortantes mediante el método de tres momentos.



$$M_1 L_1 + 2M_2(L_1 + L_2) + M_3 L_2 = -\frac{6A_1 a_1}{L_1} - \frac{6A_2 b_2}{L_2}$$

$$M_1 = -10\text{kg} * 2\text{m} * 1 = -20\text{kg} * \text{m}$$

$$\frac{6A_1 a_1}{L_1} = \frac{wL^3}{4} = \frac{10 * 4^3}{4} = 160$$

$$\frac{6A_2 b_2}{L_2} = \frac{15 * 2}{4} * (4^2 - 2^2) = 90$$

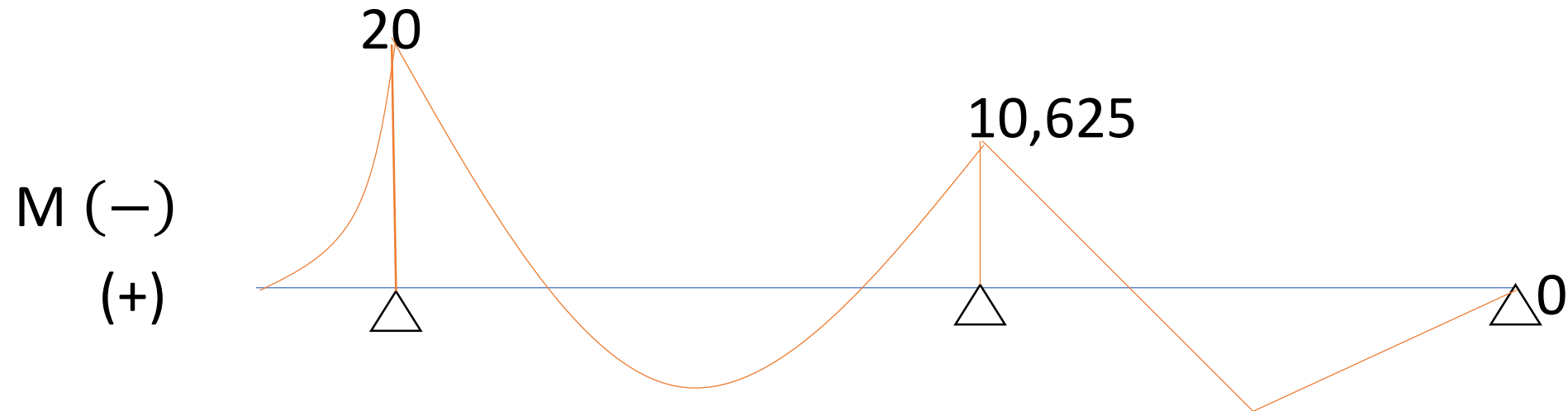
$$-20 * 4 + 2M_2(4 + 4) = -160 - 90$$

$$16M_2 = -170$$

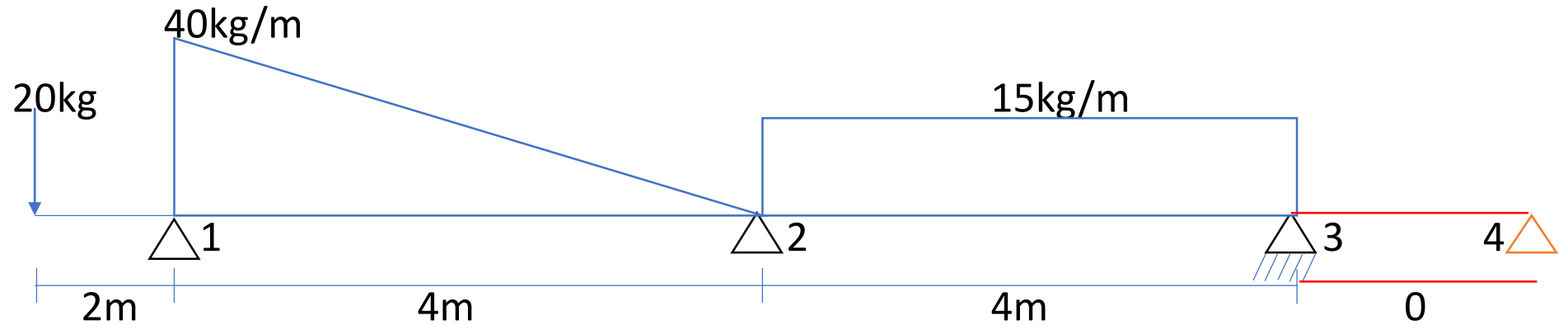
$$M_2 = -10,625$$

$$M_3 = 0$$

## DIAGRAMA DE MOMENTOS



Ejemplo 3) determinar el diagrama de momentos y cortantes mediante el método de tres momentos.



$$M_1 L_1 + 2M_2(L_1 + L_2) + M_3 L_2 = -\frac{6A_1 a_1}{L_1} - \frac{6A_2 b_2}{L_2}$$

$$M_2 L_2 + 2M_3(L_2 + L_3) + M_4 L_3 = -\frac{6A_2 a_2}{L_2} - \frac{6A_3 b_3}{L_3}$$

$$M_1 = -20kg * 2m = -40kg * m$$

$$\frac{6A_1 a_1}{L_1} = \frac{7}{60} (wL^3) = \frac{7}{60} (40 * 4^3) = 298,67$$

$$\frac{6A_2 a_2}{L_2} = \frac{6A_2 b_2}{L_2} = \frac{wL^3}{4} = \frac{15 * 4^3}{4} = 240$$

$$-40 * 4 + 2M_2(4 + 4) + M_3 * 4 = -298,67 - 240$$

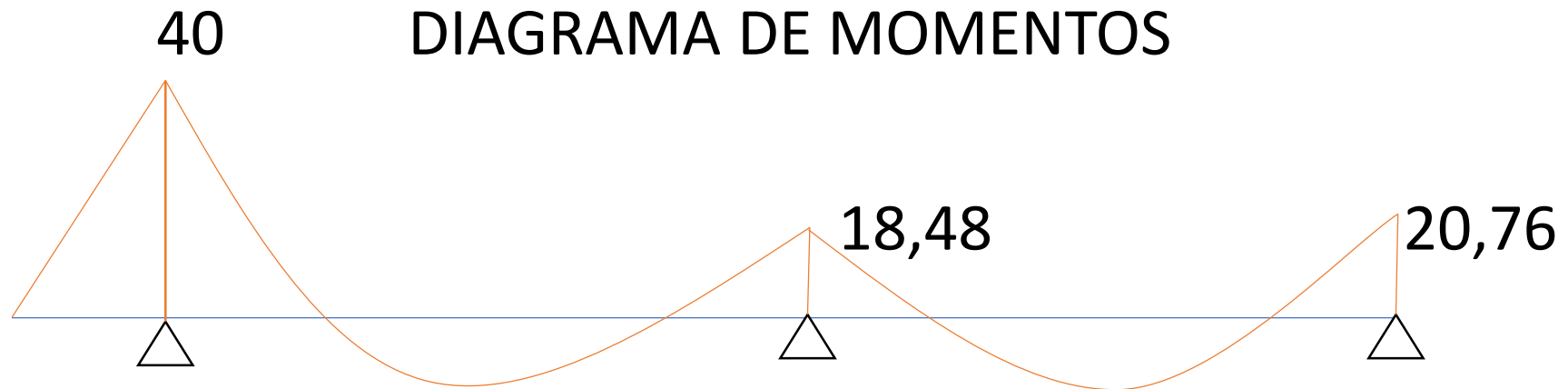
$$M_2 * 4 + 2M_3(4 + 0) + 0 = -240 - 0$$

$$16M_2 + 4M_3 = -378,67$$

$$4M_2 + 8M_3 = -240$$

$$M_2 = -18,48$$

$$M_3 = -20,76$$



$$M_1 = -40; M_2 = -18,48; M_3 = -20,76$$

DIAGRAMA DE CORTANTE

$$R_H = \frac{M_M - M_m}{L}$$

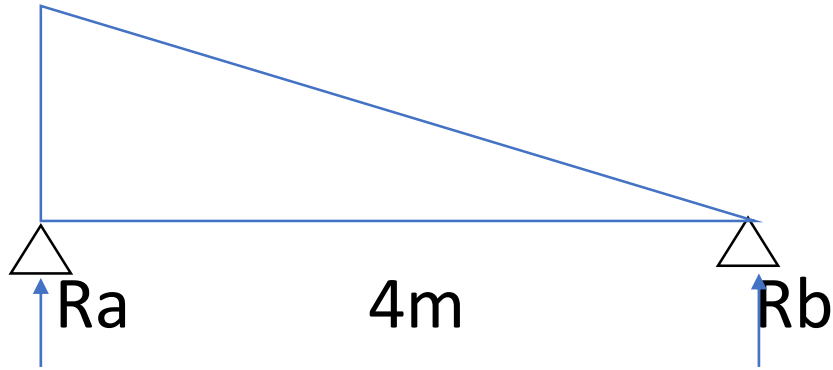
REACCION HIPERESTATICA

$$R_{H1} = \frac{40 - 18,48}{4} = 5,38$$

$$R_{H2} = \frac{20,76 - 18,48}{4} = 0,57$$

## TRAMO AB

40kg/m



$$\sum M_a = 0 + \curvearrowright$$

$$-R_b(4) + \left( \frac{40 \cdot 4}{2} * \frac{1}{3} * 4 \right) = 0$$

$$R_b = 26,67 \text{ kg}$$

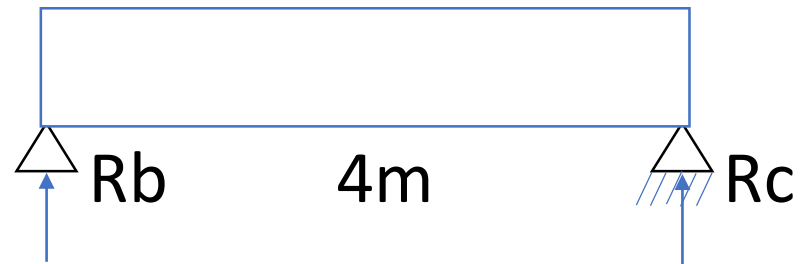
$$\sum F_v = 0 + \uparrow$$

$$R_a - \frac{40 \cdot 4}{2} + 26,67 = 0$$

$$R_a = 53,33 \text{ kg}$$

## TRAMO BC

15kg/m



$$R_b = R_c = \frac{15 \cdot 4}{2} = 30 \text{ kg}$$

$$R_T = R_I \pm R_H$$

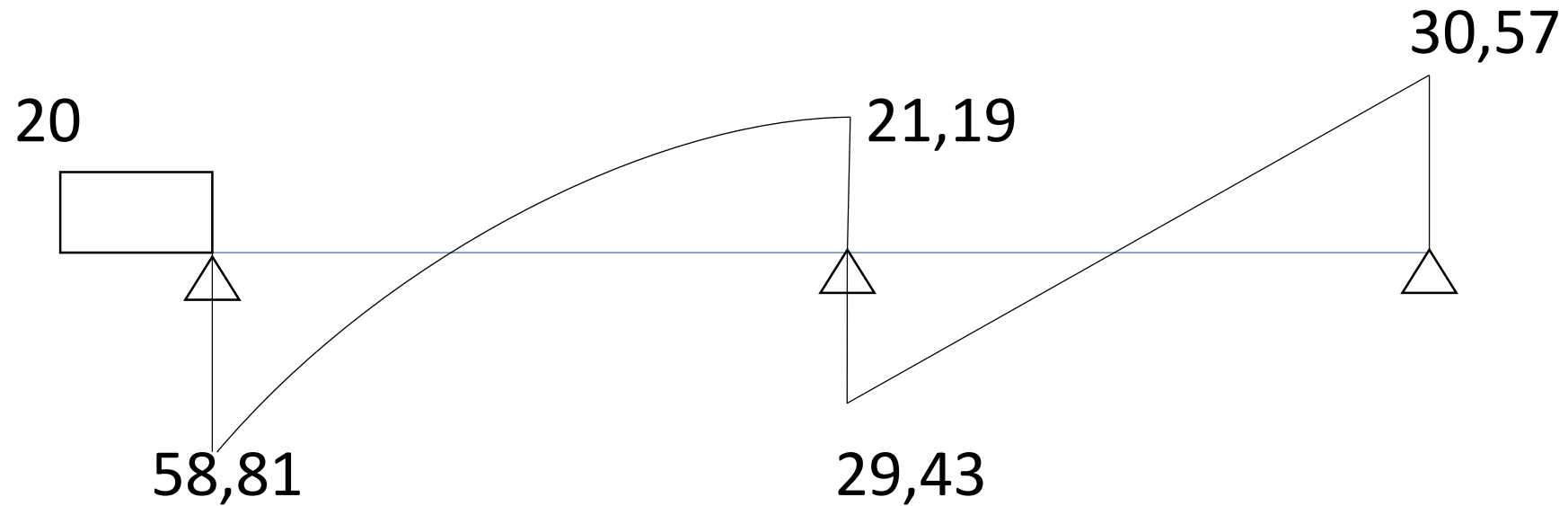
$$R_T = 53,33 + 5,38 = 58,71$$

$$R_T = 26,67 - 5,38 = 21,29$$

$$R_T = 30 - 0,57 = 29,43$$

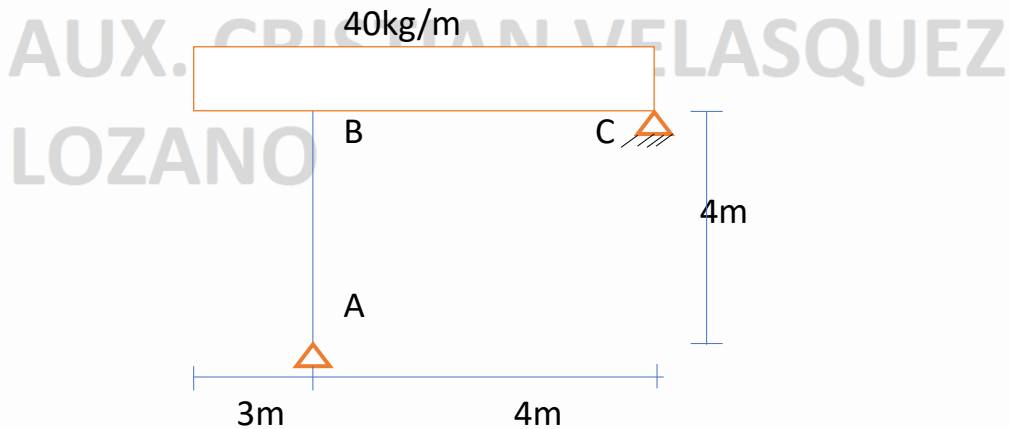
$$R_T = 30 + 0,57 = 30,57$$

### DIAGRAMA DE CORTANTES



## ROTACION ANGULAR

3) Determinar el diagrama de momentos y cortantes :C y V=2l



1) CALCULO DE MOMENTOS DE EMPOTRAMIENTO PERFECTO

$$M_{AB} = M_{BA} = 0$$

$$M_V = -40 * (3) * (1,5) = -180kg * m$$

$$M_{BC} = M_{CB} = \frac{qL^2}{12} = \frac{40 * 4^2}{12} = 53,33$$

$$M_{AB} = M_{FAB} + \frac{2EI}{L}(-2\theta_A - \theta_B)$$

$$M_{AB} = 0 + \frac{2 \cdot 1 \cdot 2}{4}(-2\theta_A - \theta_B) = -2\theta_A - \theta_B$$

$$M_{BA} = 0 + \frac{2 \cdot 1 \cdot 2}{4}(-2\theta_B - \theta_A) = -2\theta_B - \theta_A$$

$$M_{BC} = 53,33 + \frac{2 \cdot 1 \cdot 2}{4}(-2\theta_B - \theta_C) = -2\theta_B + 53,33$$

$$M_{CB} = 53,33 + \frac{2 \cdot 1 \cdot 2}{4}(-2\theta_C - \theta_B) = -\theta_B + 53,33$$

CONDICIONES

$$M_{AB} = 0 \quad : -2\theta_A - \theta_B = 0$$

$$M_{BA} + M_{BC} + M_V = 0 \quad : -2\theta_B - \theta_A - 2\theta_B + 53,33 - 180 = 0$$

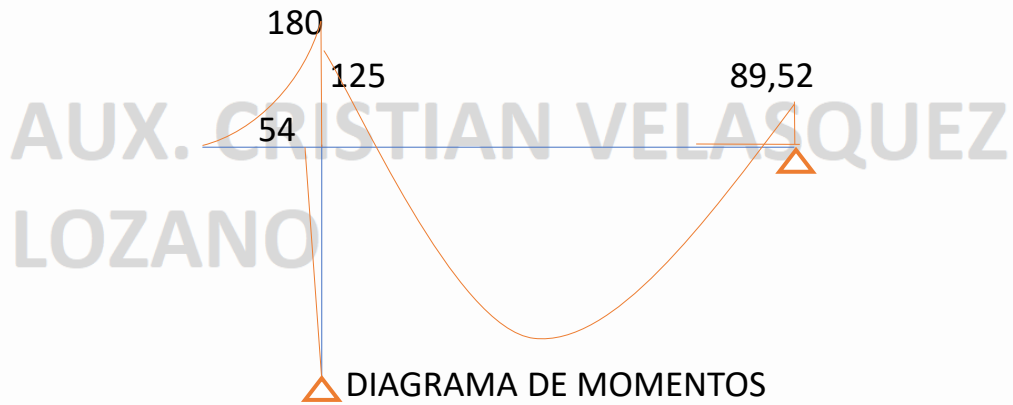
$$\theta_A = 18,1 \quad \theta_B = -36,19$$

$$M_{AB} = -2\theta_A - \theta_B = -2 \cdot (18,1) + 36,19 = 0$$

$$M_{BA} = 2 \cdot 36,19 - 18,1 = 54,28$$

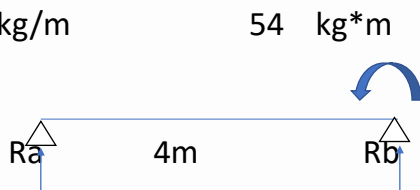
$$M_{BC} = 2 \cdot 36,19 + 53,33 = 125,71$$

$$M_{CB} = 36,19 + 53,33 = 89,52$$



TRAMO AB

40kg/m



$$\sum Ma = 0 + \curvearrowright$$

$$-Rb(4) - 54,28 = 0$$

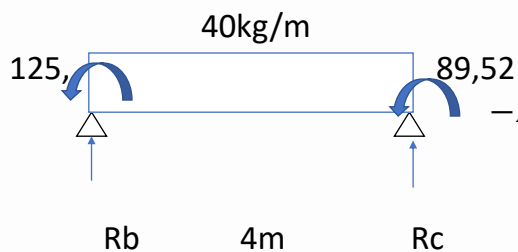
$$Rb = 13,57kg$$

$$\sum Mb = 0 + \curvearrowright$$

$$Ra * 4 - 54,28 = 0$$

$$Ra = 13,57kg$$

TRAMO BC

AUX. CRISTIAN VELASQUEZ  
LOZANO

$$\sum Mc = 0 + \curvearrowright$$

$$-Rb(4) - 40 * 4 * 2 - 125,71 - 89,52 = 0$$

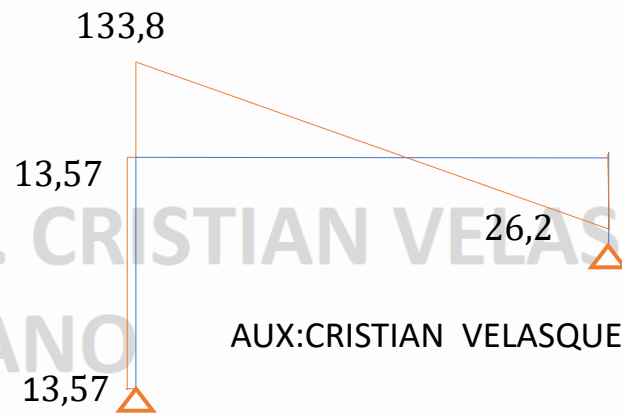
$$Rb = 133,8kg$$

$$\sum Mb = 0 + \curvearrowright$$

$$-Rc * 4 + 40 * 4 * 2 - 125,71 - 89,52 = 0$$

$$Rc = 26,2kg$$

## DIAGRAMA DE CORTANTES



AUX:CRISTIAN VELASQUEZ LOZANO

# TRES MOMENTOS

1)APOYO SIMPLE  $M=0$

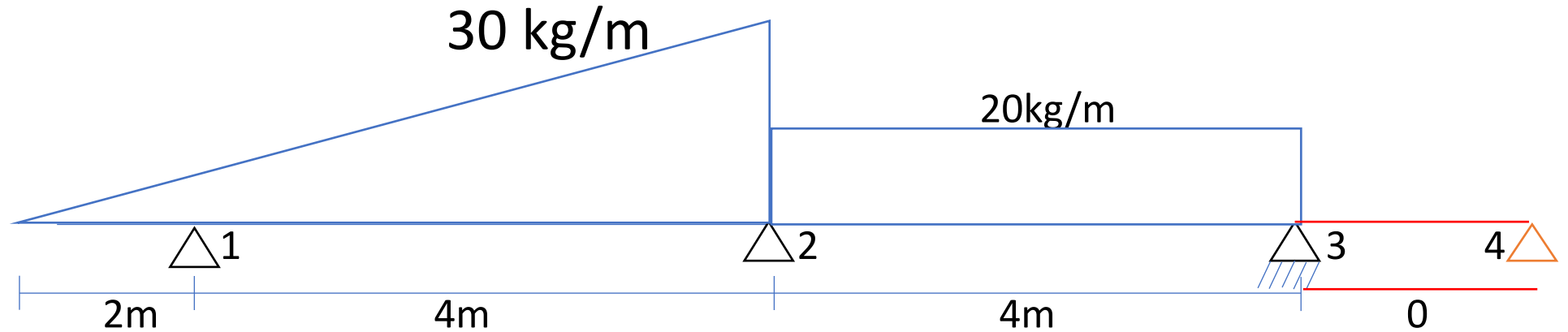
2)EMPOTRAMIENTO ALARGAMOS UN TRAMO VIRTUAL O IMAGINARIO DONDE  
TODOS LOS VALORES SON IGUALES A 0

3)VOLADIZO  $M=\text{FUERZA} \times \text{BRAZO}$

LAS TRES CONDICIONES SON PARA APOYOS EN EXTREMOS

AUXILIAR:CRISTIAN VELASQUEZ LOZANO

Ejemplo 5) determinar el diagrama de momentos y cortantes mediante el método de tres momentos.



$$M_1 L_1 + 2M_2(L_1 + L_2) + M_3 L_2 = -\frac{6A_1 a_1}{L_1} - \frac{6A_2 b_2}{L_2}$$

$$\frac{30}{6} = \frac{x}{2}$$

$$M_2 L_2 + 2M_3(L_2 + L_3) + M_4 L_3 = -\frac{6A_2 a_2}{L_2} - \frac{6A_3 b_3}{L_3}$$

$$x = 10 \text{ kg/m}$$

$$M_1 = -\frac{10 \cdot 2}{2} \cdot \frac{1}{3} \cdot 2 = -6.67 \text{ kg} \cdot \text{m}$$

$$\frac{6A_1 a_1}{L_1} = \frac{w \cdot L^3}{4} + \frac{8}{60} (w L^3) = \frac{10 \cdot 4^3}{4} + \frac{8}{60} (20 \cdot 4^3) = 330.67$$

$$\frac{6A_2 a_2}{L_2} = \frac{6A_2 b_2}{L_2} = \frac{w L^3}{4} = \frac{20 \cdot 4^3}{4} = 320$$

$$-6.67 * 4 + 2M_2(4 + 4) + M_3 * 4 = -330.67 - 320$$

$$M_2 * 4 + 2M_3(4 + 0) + 0 = -320 - 0$$

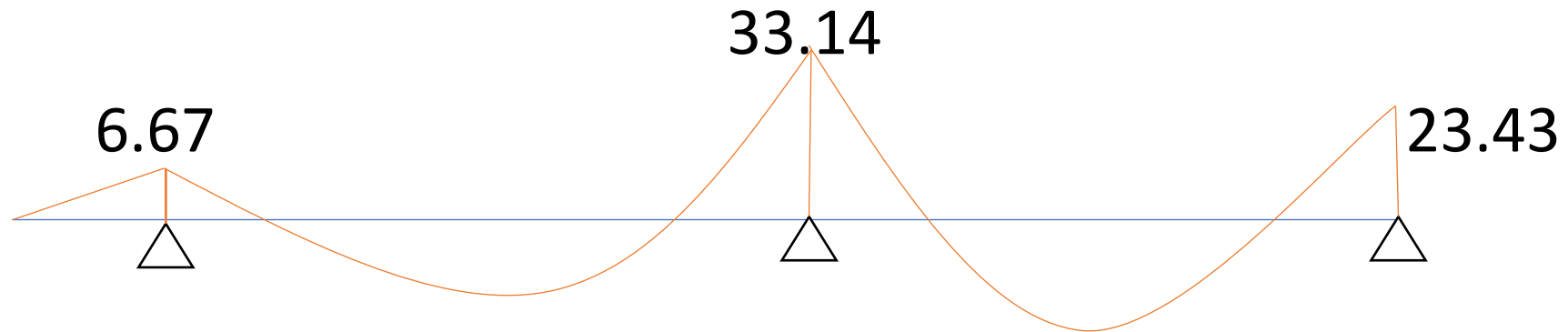
$$16M_2 + 4M_3 = -623.99$$

$$4M_2 + 8M_3 = -320$$

$$M_2 = -33.14 \text{ kg} * \text{m}$$

$$M_3 = -23.43 \text{ kg} * \text{m}$$

### DIAGRAMA DE MOMENTOS



$$M_1 = -6,67; M_2 = -33,14; M_3 = -23,43$$

DIAGRAMA DE CORTANTE

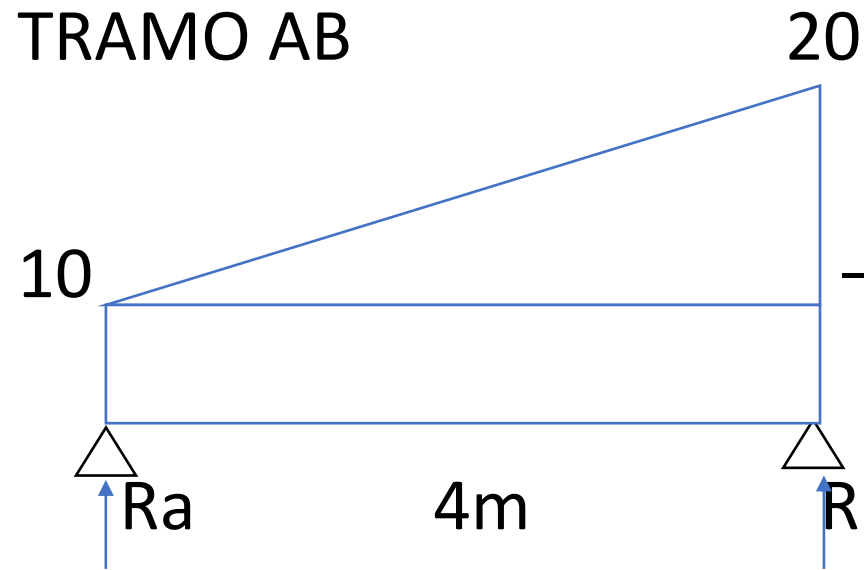
$$R_H = \frac{M_M - M_m}{L}$$

REACCION HIPERESTATICA

$$R_{H1} = \frac{33.14 - 6.67}{4} = 6.62$$

$$R_{H2} = \frac{33.14 - 23.43}{4} = 2.43$$

TRAMO AB



$$\sum Ma = 0 + \curvearrowright$$

$$-Rb(4) + \left( \frac{20 \cdot 4}{2} \cdot \frac{2}{3} \cdot 4 \right) + (10 \cdot 4 \cdot 2) = 0$$

$$Rb = 46.67 \text{ kg}$$

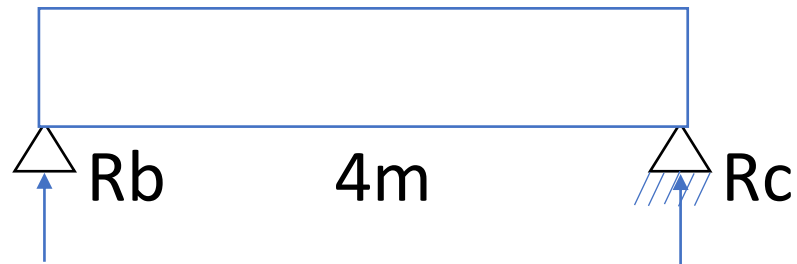
$$\sum Fv = 0 + \uparrow$$

$$Ra - \frac{20 \cdot 4}{2} - (10 \cdot 4) + 46.67 = 0$$

$$Ra = 33.33 \text{ kg}$$

TRAMO BC

20 kg/m



$$Rb = Rc = \frac{20 \cdot 4}{2} = 40 \text{ kg}$$

$$R_T = R_I \pm R_H$$

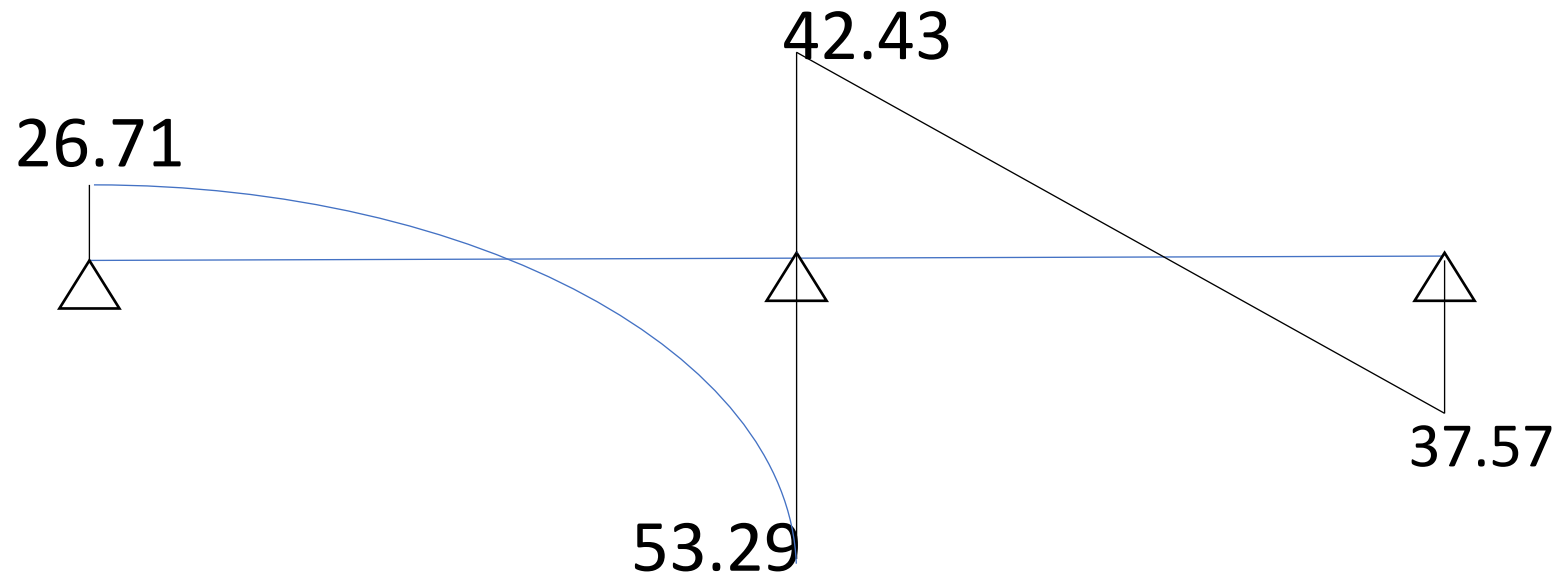
$$R_T = 33,33 - 6,62 = 26,71$$

$$R_T = 46,67 + 6,62 = 53,29$$

$$R_T = 40 + 2,43 = 42,43$$

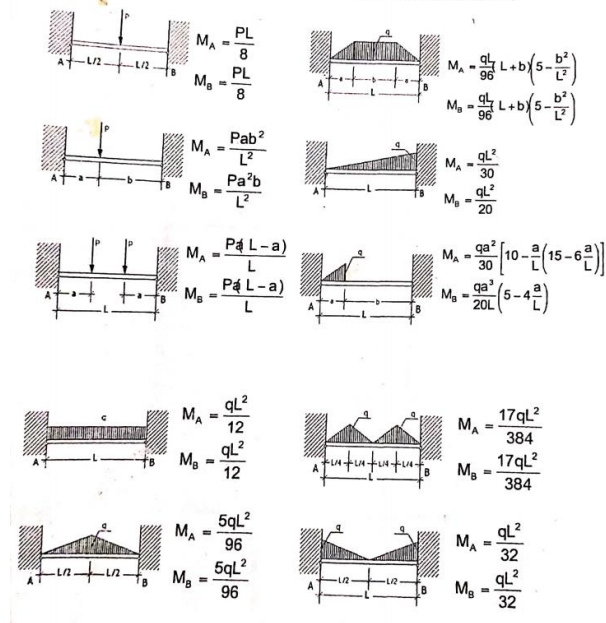
$$R_T = 40 - 2,43 = 37,57$$

### DIAGRAMA DE CORTANTES



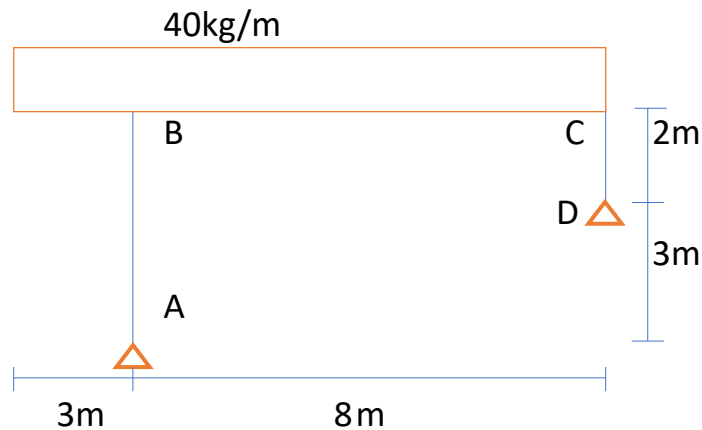
Estructuras Hiperestáticas

3.8 Tabla de momentos de empotramientos en los extremos



## HARDY CROSS

3) Determinar el diagrama de momentos y cortantes



## 1) CALCULO DE MOMENTOS DE EMPOTRAMIENTO PERFECTO

$$M_{AB} = M_{BA} = 0 \qquad M_{CD} = M_{DC} = 0$$

$$M_V = -40 * (3) * (1,5) = -180 \text{ kg} * \text{m}$$

$$M_{BC} = M_{CB} = \frac{qL^2}{12} = \frac{40 * 8^2}{12} = 213,33$$

## 2) FACTOR DE DISTRIBUCION

$$FD_{AB} = 1$$

$$FD_{DC} = 1$$

$$FD = \frac{K}{\sum K} ; K = \frac{\text{Inercia}}{\text{Longitud}}$$

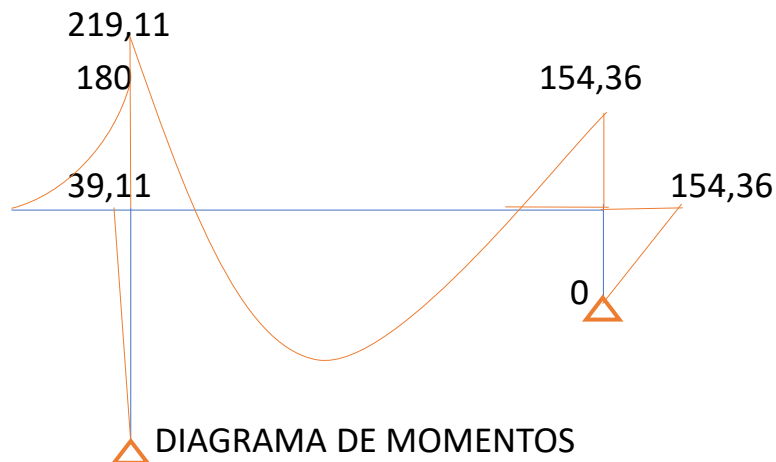
$$FD_{BA} = \frac{\frac{1}{5}}{\frac{1}{5} + \frac{1}{8}} = 0,62$$

$$FD_{BC} = \frac{\frac{1}{8}}{\frac{1}{8} + \frac{1}{5}} = 0,38$$

$$FD_{CB} = \frac{\frac{1}{8}}{\frac{1}{8} + \frac{1}{3}} = 0,27$$

$$FD_{BC} = \frac{\frac{1}{3}}{\frac{1}{3} + \frac{1}{8}} = 0,73$$

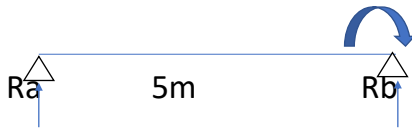
| NUDO  | A      | B      |         |        | C       |        | D      |
|-------|--------|--------|---------|--------|---------|--------|--------|
| TRAMO | AB     | BA     | V       | BC     | CB      | CD     | DC     |
| F.D.  | 1,00   | 0,62   | 0,00    | 0,38   | 0,27    | 0,73   | 1,00   |
| M.E.P | 0,00   | 0,00   | -180,00 | 213,33 | -213,33 | 0,00   | 0,00   |
| 1 I   | 0,00   | -20,66 | 0,00    | -12,67 | 57,60   | 155,73 | 0,00   |
|       | -10,33 | 0,00   | 0,00    | 28,80  | -6,33   | 0,00   | 77,87  |
| 2 I   | 10,33  | -17,86 | 0,00    | -10,94 | 1,71    | 4,62   | -77,87 |
|       | -8,93  | 5,17   | 0,00    | 0,85   | -5,47   | -38,94 | 2,31   |
| 3 I   | 8,93   | -3,73  | 0,00    | -2,29  | 11,99   | 32,42  | -2,31  |
|       | -1,87  | 4,46   | 0,00    | 5,99   | -1,14   | -1,16  | 16,21  |
| 4 I   | 1,87   | -6,48  | 0,00    | -3,97  | 0,62    | 1,68   | -16,21 |
|       | 0,00   | -39,11 | -180,00 | 219,11 | -154,36 | 154,36 | 0,00   |



TRAMO AB

40kg/m

39,11kg\*m



$$\sum M_a = 0 + \curvearrowright$$

$$-R_b(5) + 39,11 = 0$$

$$R_b = 7,822kg$$

$$\sum M_b = 0 + \curvearrowright$$

$$R_a * 5 + 39,11 = 0$$

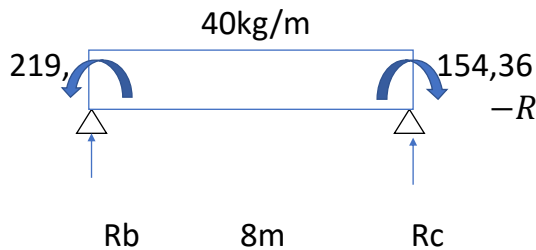
$$R_a = -7,822kg$$

TRAMO BC

$$\sum M_c = 0 + \curvearrowright$$

$$-R_b(8) - 40 * 8 * 4 + 154,36 - 219,11 = 0$$

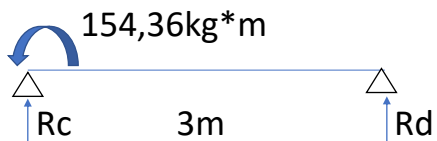
$$R_b = 168,1kg$$



$$\sum M_b = 0 + \curvearrowright$$

$$-R_c * 8 + 40 * 8 * 4 + 154,36 - 219,11 = 0$$

$$R_c = 151,9kg$$



$$\sum c = 0 +$$

$$-R_d(3) - 154,36 = 0$$

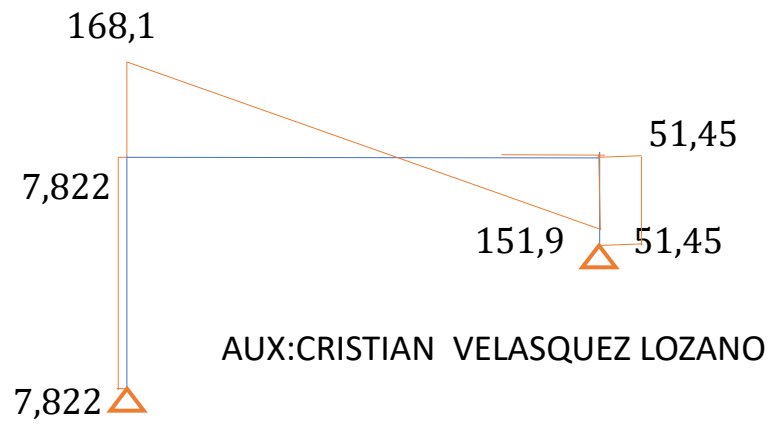
$$R_d = -51,45kg$$

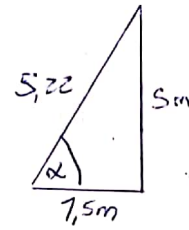
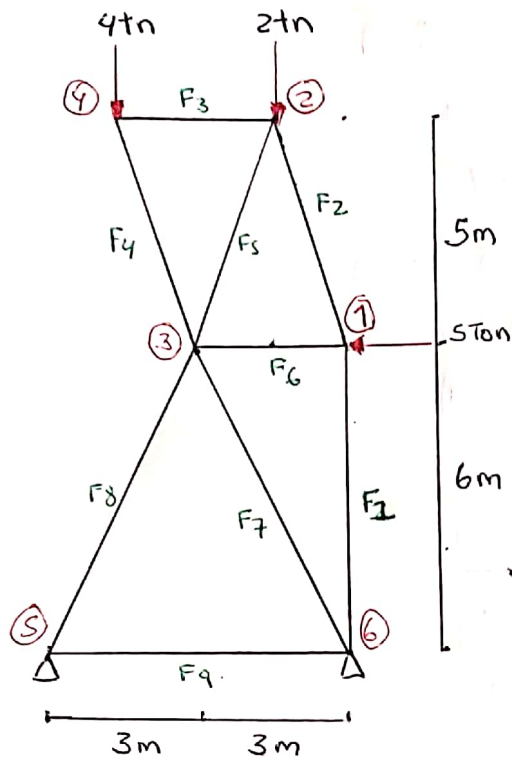
$$\sum M_d = 0 +$$

$$R_c * 3 - 154,36 = 0$$

$$R_c = 51,45kg$$

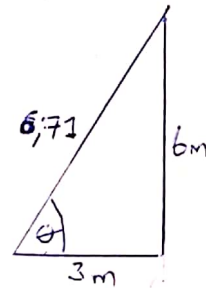
## DIAGRAMA DE CORTANTES





$$\sin \alpha = \frac{5}{5,22} = 0,96 (y)$$

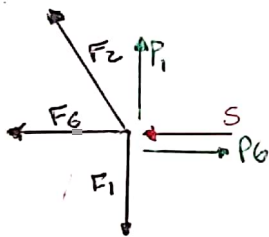
$$\cos \alpha = \frac{1,5}{5,22} = 0,29 (x)$$



$$\sin \theta = \frac{6}{6,71} = 0,89 (y)$$

$$\cos \theta = \frac{3}{6,71} = 0,45 (x)$$

NUDO 1 D.C.L



$$\sum F_y = 0 \uparrow (+)$$

$$P_1 - F_1 + F_2(y) = 0$$

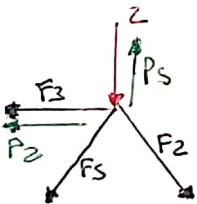
$$P_1 = F_1 + 0,96(F_2)$$

$$\sum F_x = 0 \rightarrow (+)$$

$$P_6 - F_6 - S - F_2(x) = 0$$

$$P_6 = F_2(0,29) + F_6 + S$$

NUDO 2 D.C.L



$$\sum F_y = 0 \uparrow (+)$$

$$P_2 - F_2(y) - F_5(y) = 0$$

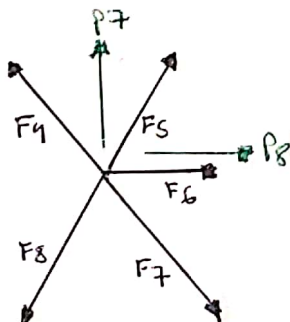
$$P_2 = F_2(0,96) + F_5(0,96) + Z$$

$$\sum F_x = 0 \rightarrow (+)$$

$$-P_3 - F_3 - F_5(x) + F_2(x) = 0$$

$$P_3 = F_2(0,29) - F_3 - F_5(0,29)$$

NUDO 3 D.C.L



$$\sum F_y = 0 \uparrow (+)$$

$$P_7 + F_5(y) + F_4(y) - F_7(y) - F_8(y) = 0$$

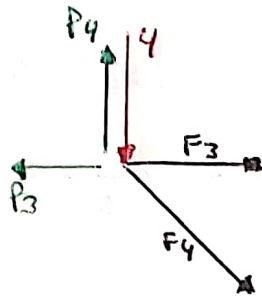
$$P_7 = -F_4(0,96) - F_5(0,96) + F_7(0,89) + F_8(0,89)$$

$$\sum F_x = 0 \rightarrow (+)$$

$$P_8 + F_6 + F_5(x) - F_4(x) + F_7(x) - F_8(x) = 0$$

$$P_8 = F_4(0,29) - F_5(0,29) - F_6 - F_7(0,45) + F_8(0,45)$$

NUDO 4



$$\sum F_y = 0 \uparrow (+)$$

$$P_4 - 4 - F_4(y) = 0$$

$$P_4 = F_4(0,96) + 4$$

$$\sum F_x = 0 \rightarrow (+)$$

$$-P_3 + F_3 + F_4(x) = 0$$

$$P_3 = F_3 + F_4(0,29)$$

| $F_1$ | $F_2$ | $F_3$ | $F_4$ | $F_5$ | $F_6$ | $F_7$ | $F_8$ |     |
|-------|-------|-------|-------|-------|-------|-------|-------|-----|
| 1     | 0,96  | 0     | 0     | 0     | 0     | 0     | 0     | = 0 |
| 0     | 0,29  | -1    | 0     | -0,29 | 0     | 0     | 0     | = 0 |
| 0     | 0     | 1     | 0,29  | 0     | 0     | 0     | 0     | = 0 |
| 0     | 0     | 0     | 0,96  | 0     | 0     | 0     | 0     | = 4 |
| 0     | 0,96  | 0     | 0     | 0,96  | 0     | 0     | 0     | = 2 |
| 0     | 0,29  | 0     | 0     | 0     | 1     | 0     | 0     | = 5 |
| 0     | 0     | 0     | -0,96 | -0,96 | 0     | 0,89  | 0,89  | = 0 |
| 0     | 0     | 0     | 0,29  | -0,29 | -1    | -0,45 | 0,45  | = 0 |

$$F_1 = 1$$

$$F_2 = -1,04$$

$$F_3 = -1,20$$

$$F_4 = 4,17$$

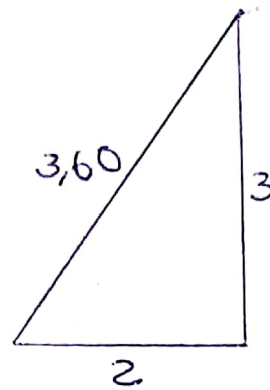
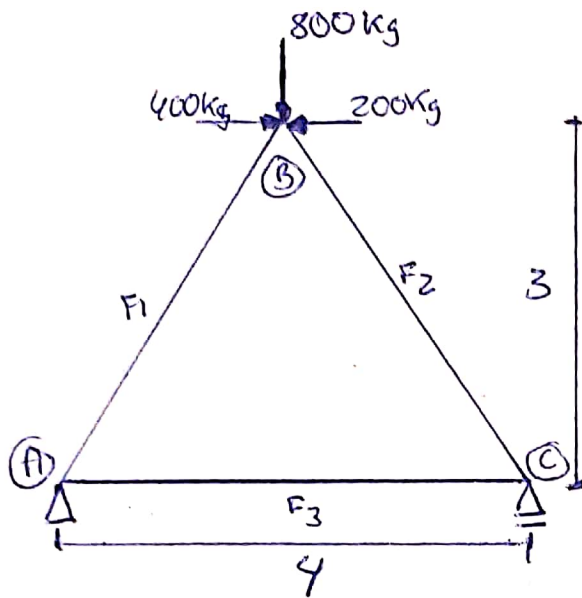
$$F_5 = 3,13$$

$$F_6 = 5,30$$

$$F_7 = 1,6$$

$$F_8 = 9,5$$

$$F_9 = 0$$



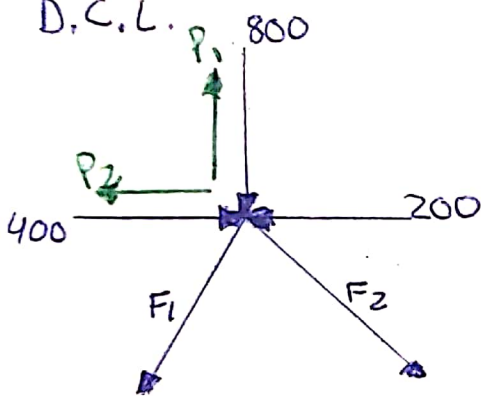
1) PITAGORAS  
 $H = \sqrt{2^2 + 3^2}$   
 $H = 3,60$

$$\sin \alpha = \frac{3}{3,60} = 0,83 \text{ (y)}$$

$$\cos \alpha = \frac{2}{3,60} = 0,55 \text{ (x)}$$

NUDO B

D.C.L.



$$\sum F_V = 0 \quad \uparrow (+)$$

$$P_1 - 800 - F_1 y - F_2 y = 0$$

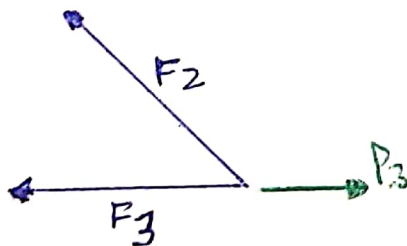
$$P_1 = F_1(0,83) + F_2(0,83) + 800$$

$$\sum F_H = 0 \quad \rightarrow (+)$$

$$-P_2 - F_1(x) + F_2(x) - 200 + 400 = 0$$

$$P_2 = -F_1(0,55) + F_2(0,55) + 200$$

NUDO C D.C.L



$$\sum F_H = 0 \quad \rightarrow (+)$$

$$P_3 - F_3 - F_2(x) = 0$$

$$P_3 = F_2(0,55) + F_3$$

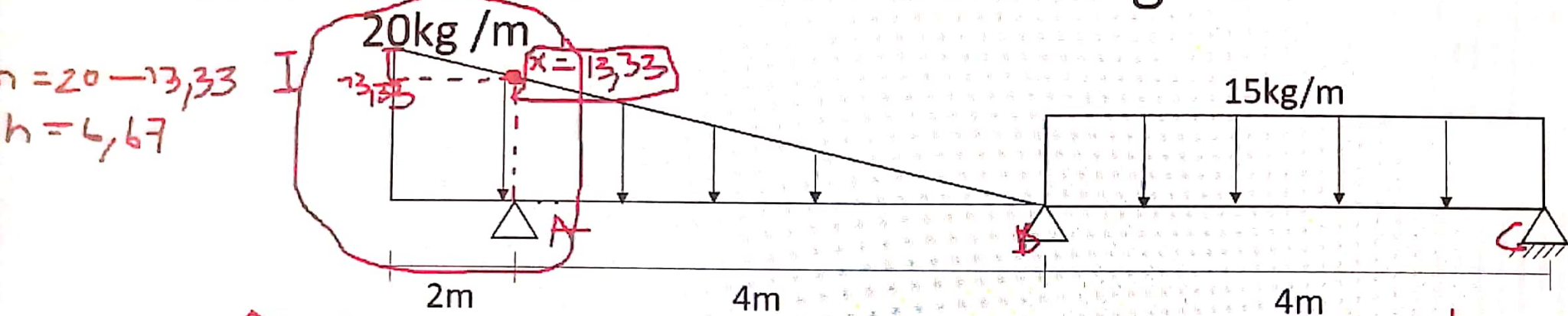
|         |          |       |         |
|---------|----------|-------|---------|
| $F_1$   | $F_2$    | $F_3$ |         |
| $0,83$  | $+ 0,83$ | $+ 0$ | $= 800$ |
| $-0,55$ | $+ 0,55$ | $+ 0$ | $= 200$ |
| $0$     | $+ 0,55$ | $+ 1$ | $= 0$   |

$$F_1 = 300,1 \text{ (TENSION)}$$

$$F_2 = 663,7 \text{ (TENSION)}$$

$$F_3 = -365,1 \text{ (COMPRESION)}$$

Ejemplo 2) determinar el diagrama de momentos mediante el método de rotación angular.



Se calcula el momento fijo en B para la viga AB:

$$M_v = \left( -13,33 \cdot 2 \cdot \frac{1}{3} \right) + \left( -\frac{6,67 \cdot 2}{2} \cdot \frac{2}{3} \cdot 2 \right) = -35,55 \text{ kg} \cdot \text{m}$$

$$M_{AB} = \frac{qL^2}{20} = \frac{13,33 \cdot 4^2}{20} = 10,66 \text{ F}$$

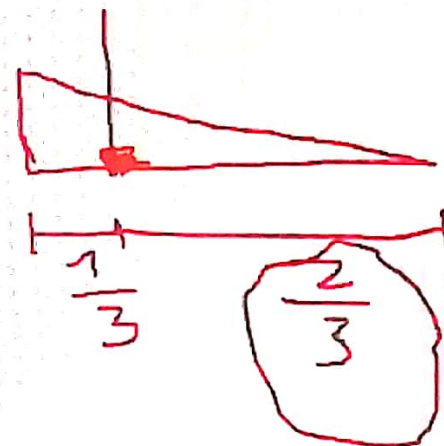
Se calculan los momentos fijos en B para las vigas AB y BC:

$$M_{BA} = \frac{qL^2}{30} = \frac{13,33 \cdot 4^2}{30} = 7,11$$

$$M_{BC} = \frac{qL^2}{12} = \frac{15 \cdot 4^2}{12} = 20$$

Se aplica la ecuación de equilibrio de momentos en B:

$$M_{AB} = M_{FAB} + \frac{2EI}{L} (-2\theta_A - \theta_B)$$



$$M_{AB} = M_{FAB} + \frac{2EI}{L}(-2\theta_A - \theta_B)$$

$$M_{AB} = 10,66 + \frac{2 \cdot 1 \cdot 1}{4}(-2\theta_A - \theta_B) = -\theta_A - 0,5\theta_B + 10,66$$

$$M_{BA} = 7,11 + \frac{2 \cdot 1 \cdot 1}{4}(-2\theta_B - \theta_A) = -\theta_B - 0,5\theta_A + 7,11$$

$$M_{BC} = 20 + \frac{2 \cdot 1 \cdot 1}{4}(-2\theta_B - \theta_C) = -\theta_B + 20$$

$$M_{CB} = 20 + \frac{2 \cdot 1 \cdot 1}{4}(-2\theta_C - \theta_B) = -0,5\theta_B + 20$$

$$E = 1$$

$$I = 1$$

### CONDICIONES

$$M_{AB} + M_V = 0 : \overset{M_{AB}}{-\theta_A - 0,5\theta_B + 10,66} - \overset{M_V}{35,55} = 0$$

$$M_{BA} + M_{BC} = 0 : \overset{M_{BA}}{-\theta_B - 0,5\theta_A + 7,11} - \overset{M_{BC}}{\theta_B + 20} = 0$$

$M \rightarrow$  FISO = 0  
O MOVIL



$$\theta_A = -36,19$$

$$\theta_B = 22,60$$

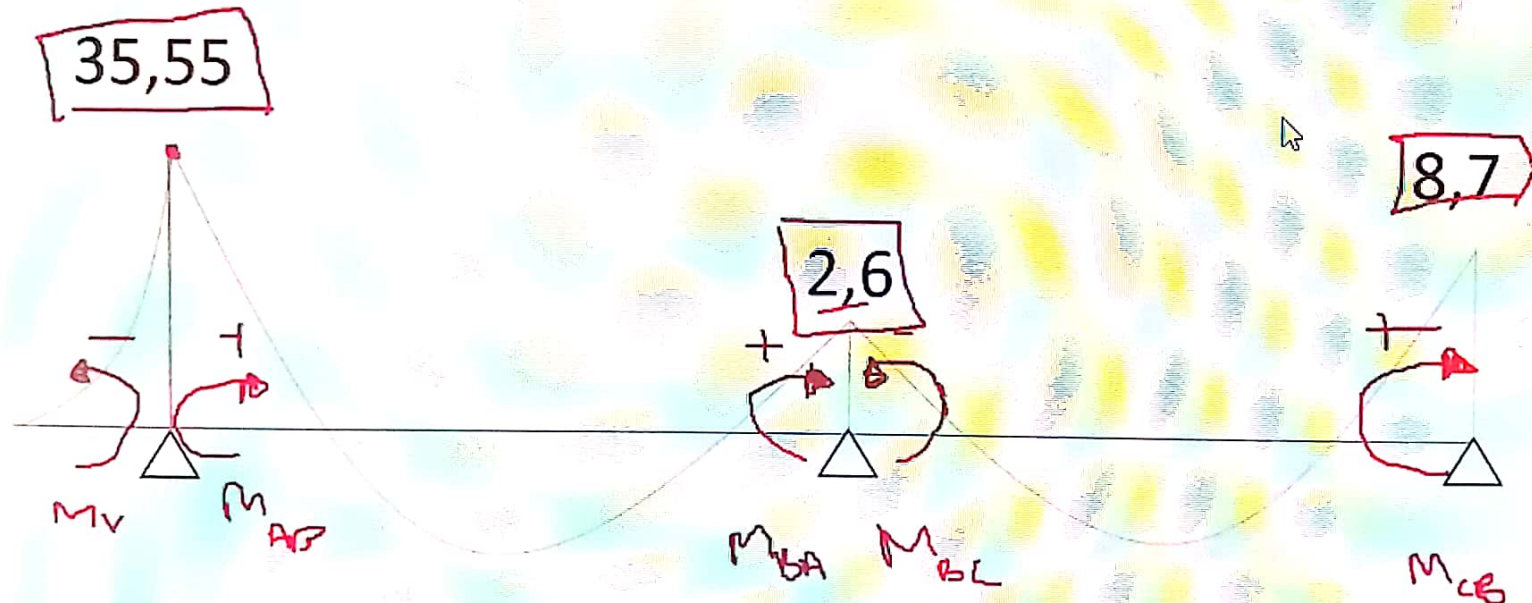
$$M_{AB} = -\theta_A - 0,5\theta_B + 10,66 = 36,19 - 0,5 * 22,60 + 10,66 = 35,55$$

$$M_{BA} = -22,60 - 0,5 * (-36,19) + 7,11 = 2,605$$

$$M_{BC} = -22,60 + 20 = -2,6$$

$$M_{CB} = -0,5 * 22,60 + 20 = 8,7$$

### DIAGRAMA DE MOMENTOS



# 1) CALCULO DE MOMENTOS DE EMPOTRAMIENTO PERFECTO

$$M_{AB} = M_{BA} = 0$$

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$$M_V = \underbrace{-40 * (3)}_F * \underbrace{(1,5)}_{\downarrow} = -180 \text{ kg} * \text{m}$$

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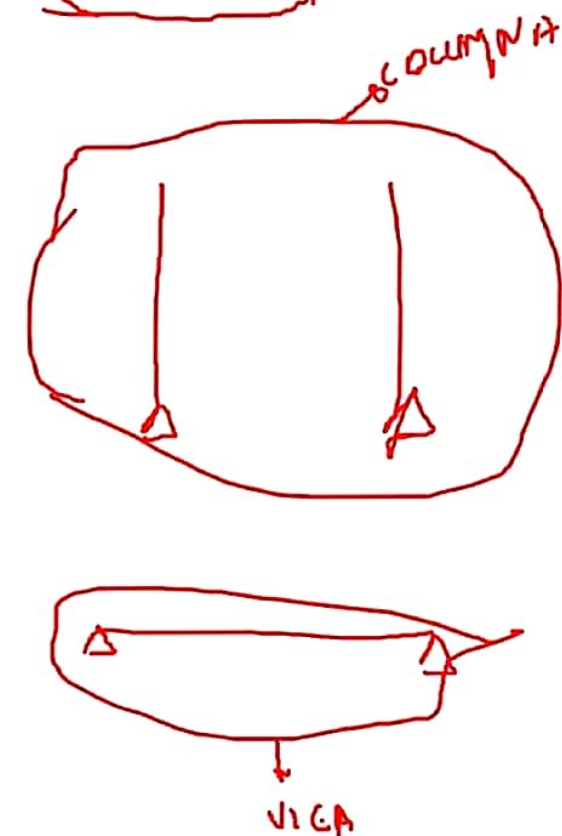
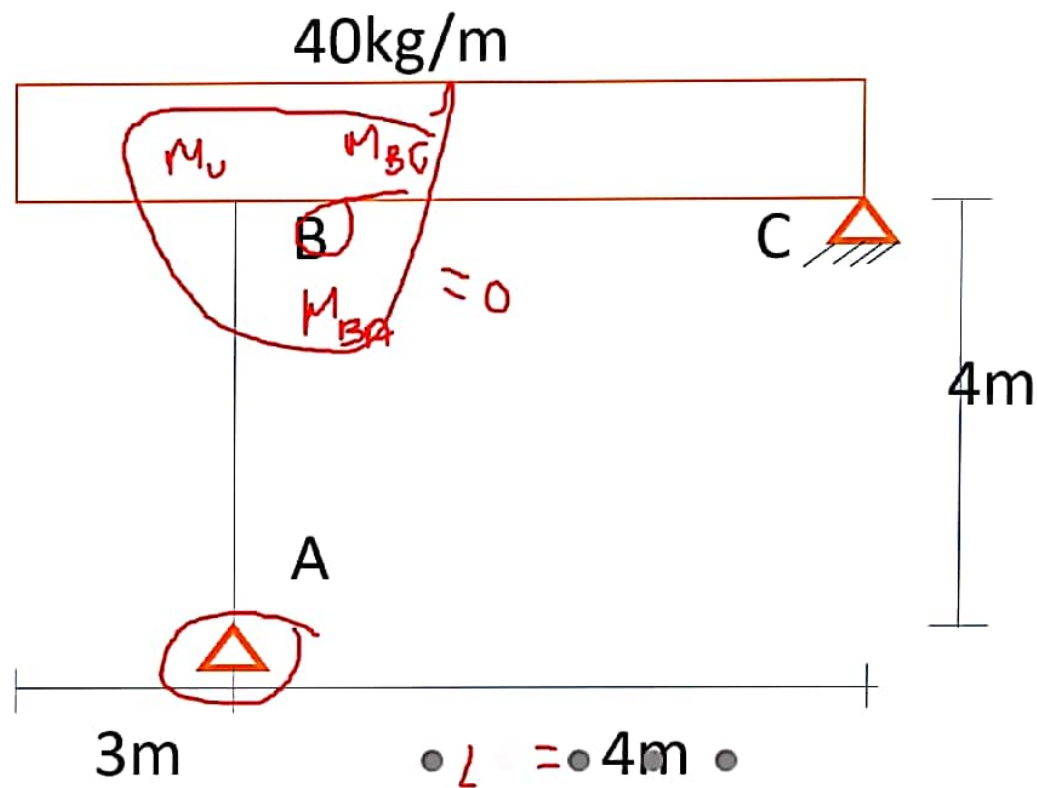
$$M_{BC} = M_{CB} = \frac{qL^2}{12} = \frac{40 * 4^2}{12} = 53,33$$

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# ROTACION ANGULAR

3) Determinar el diagrama de momentos y cortantes :C y V=I=2



$$M_{AB} = M_{FAB} + \frac{2EI}{L}(-2\theta_A - \theta_B)$$

$$M_{AB} = 0 + \frac{2 \cdot 1 \cdot 2}{4}(-2\theta_A - \theta_B) = -2\theta_A - \theta_B$$

$$M_{BA} = 0 + \frac{2 \cdot 1 \cdot 2}{4}(-2\theta_B - \theta_A) = -2\theta_B - \theta_A$$

$$M_{BC} = 53,33 + \frac{2 \cdot 1 \cdot 2}{4}(-2\theta_B - \theta_C) = -2\theta_B + 53,33$$

$$M_{CB} = 53,33 + \frac{2 \cdot 1 \cdot 2}{4}(-2\theta_C - \theta_B) = -\theta_B + 53,33$$

CONDICIONES

$$M_{AB} = 0$$

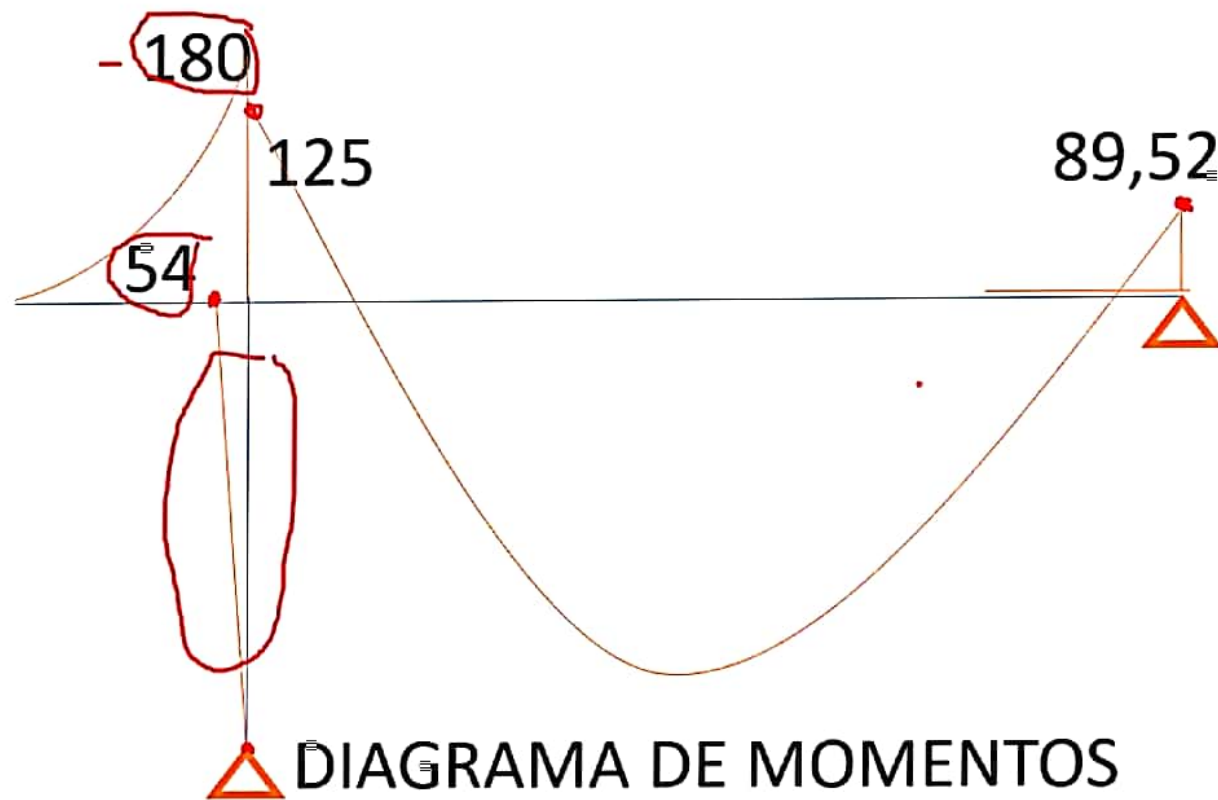
$$: -2\theta_A - \theta_B = 0$$

$$M_{BA} + M_{BC} + M_V = 0$$

$$: -2\theta_B - \theta_A - 2\theta_B + 53,33 - 180 = 0$$



M (-)  
(+)



$$\theta_A = 18,1$$

$$\theta_B = -36,19$$

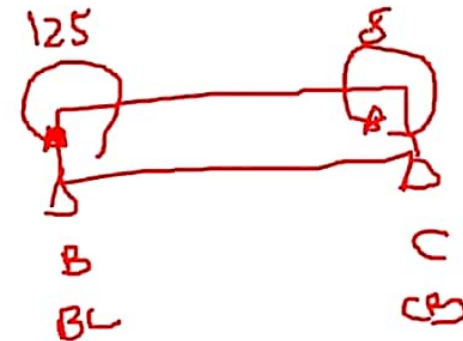
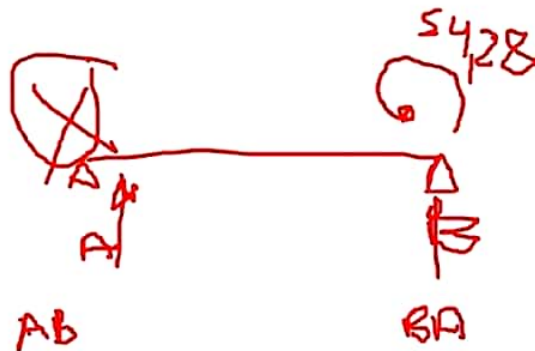
$$\sum M \Delta = 0$$

$$M_{AB} = -2\theta_A - \theta_B = -2*(18,1) + 36,19 = 0$$

$$M_{BA} = 2 * 36,19 - 18,1 = 54,28$$

$$M_{BC} = 2 * 36,19 + 53,33 = 125,71$$

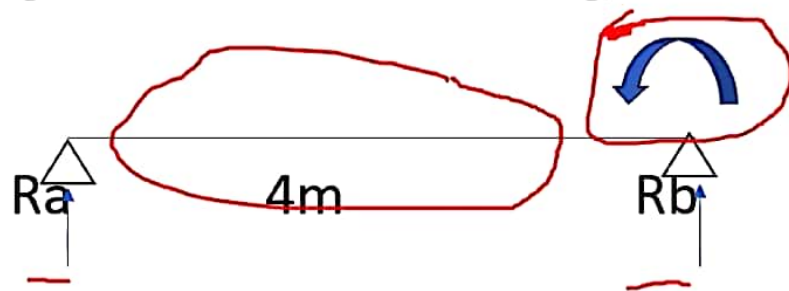
$$M_{CB} = 36,19 + 53,33 = 89,52$$



TRAMO AB

40kg/m

54 kg\*m



$$\sum Ma = 0 + \curvearrowright$$

$$-Rb(4) - 54,28 = 0$$

$$Rb = 13,57kg$$

$$\sum Mb = 0 + \curvearrowright$$

$$Ra * 4 - 54,28 = 0$$

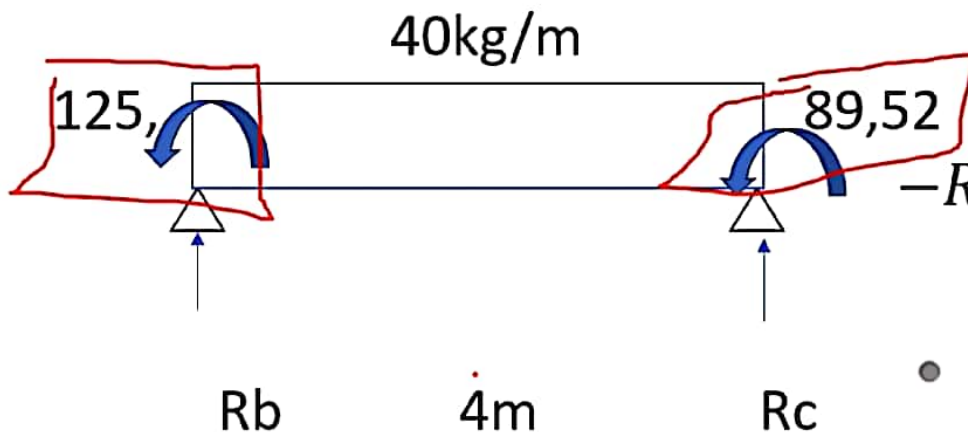
$$Ra = 13,57kg$$

TRAMO BC

$$\sum Mc = 0 + \curvearrowright$$

$$-Rb(4) - 40 * 4 * 2 - 125,71 - 89,52 = 0$$

$$Rb = 133,8kg$$



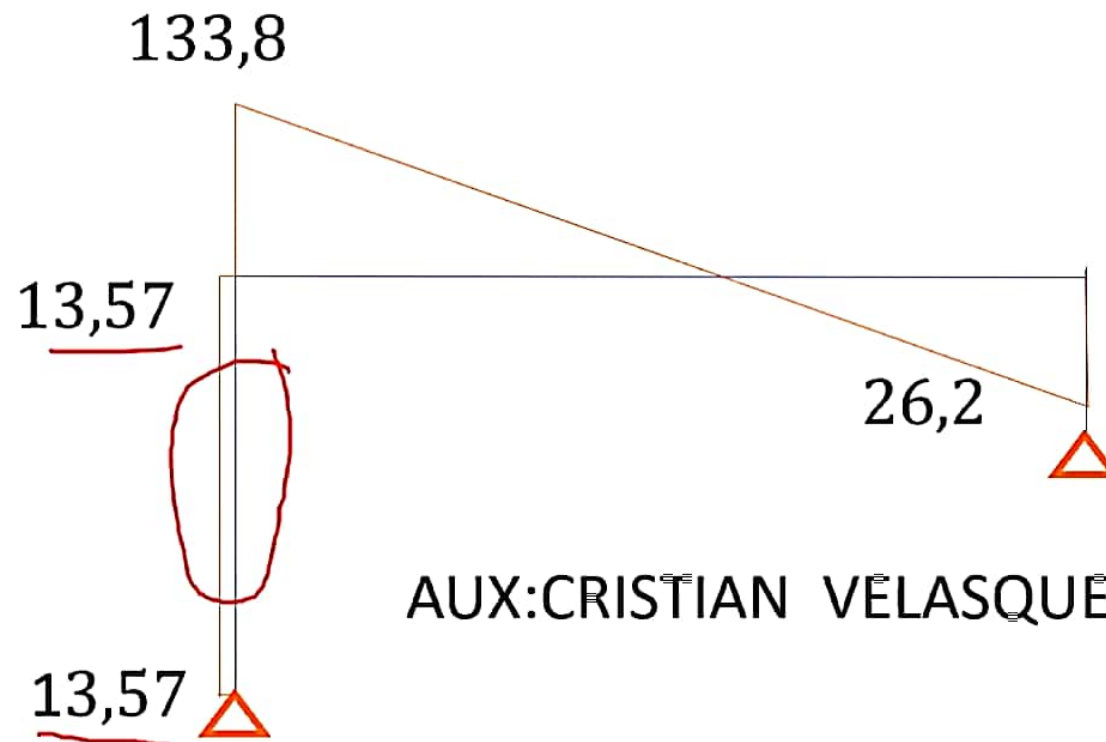
$$\sum Mb = 0 + \curvearrowright$$

$$-Rc * 4 + 40 * 4 * 2 - 125,71 - 89,52 = 0$$

$$Rc = 26,2kg$$

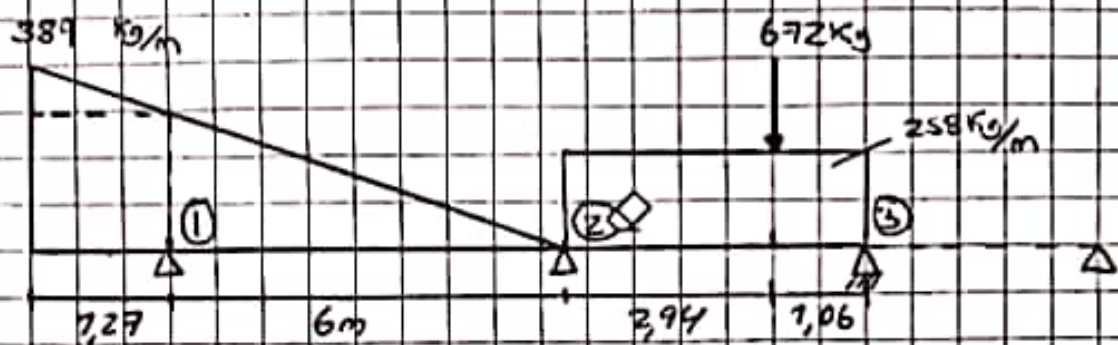


# DIAGRAMA DE CORTANTES



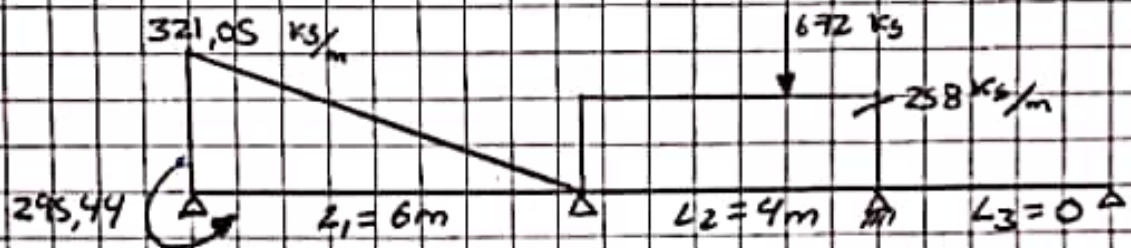
AUX:CRISTIAN VELASQUEZ LOZANO





$$\frac{389}{7.27} = \frac{x}{6} \quad ; \quad x = 321.05$$

$$M_1 = - \left( 321.05 (1.27) (0.635) + \frac{67.95 (1.27)}{2} + \frac{2}{3} \cdot (1.27) \right) = -295.44 \text{ Kg}\cdot\text{m}$$



$$M_1(L_1) + 2 M_2(L_1 + L_2) + M_3 L_2 = - \frac{6 A_1 a_1}{L_1} - \frac{6 A_2 b_2}{L_2}$$

$$M_2(L_2) + 2 M_3(L_2 + L_3) + M_4 L_3 = - \frac{6 A_2 a_2}{L_2} - \frac{6 A_3 b_3}{L_3}$$

$$\frac{6 A_1 a_1}{L_1} = \frac{7}{60} (w) (L_1)^3 = \frac{7}{60} (321) (6)^3 = 8089.2$$

$$\frac{6 A_2 b_2}{L_2} = \frac{w L^3}{4} + \frac{P b}{L} (L^2 - b^2) = \frac{258 (4)^3}{4} + \frac{672 (1.06)}{4} (4^2 - 1.06^2) = 6777.19$$

$$\frac{6 A_2 a_2}{L_2} = \frac{w L^3}{4} + \frac{P a}{L} (L^2 - a^2) = \frac{258 (4)^3}{4} + \frac{672 (2.94)}{4} (4^2 - 2.94^2) = 7767.47$$

$$-295.44 (6) + 2 M_2 (6 + 4) + M_3 (4) = -8089.2 - 6777.19$$

$$M_2 (4) + 2 M_3 (4) = -7771.47$$

$$\begin{cases} 20 (M_2) + 4 (M_3) = -7771.47 \end{cases}$$

$$M_2(L_2) + 2M_3(L_2 + L_3) + M_4L_3 = -\frac{6H_2a_2}{L_2} - \frac{6H_3b_3}{L_3}$$

$$\frac{6A_1a_1}{L_1} = \frac{7}{60}(w)(L)^3 = \frac{7}{60}(327)(6)^3 = 8089,2$$

$$\frac{6A_2b_2}{L_2} = \frac{wL^3}{4} + \frac{Pb}{L}(L^2 - b^2) = \frac{258(4)^3}{4} + \frac{672(7,06)}{4}(4^2 - 1,06^2) = 6777,19$$

$$\frac{6A_2a_2}{L_2} = \frac{wL^3}{4} + \frac{Pa}{L}(L^2 - a^2) = \frac{258(4)^3}{4} + \frac{672(2,94)}{4}(4^2 - 2,94^2) = 7767,47$$

$$-295,44(6) + 2M_2(6+4) + M_3(4) = -8089,2 - 6777,19$$

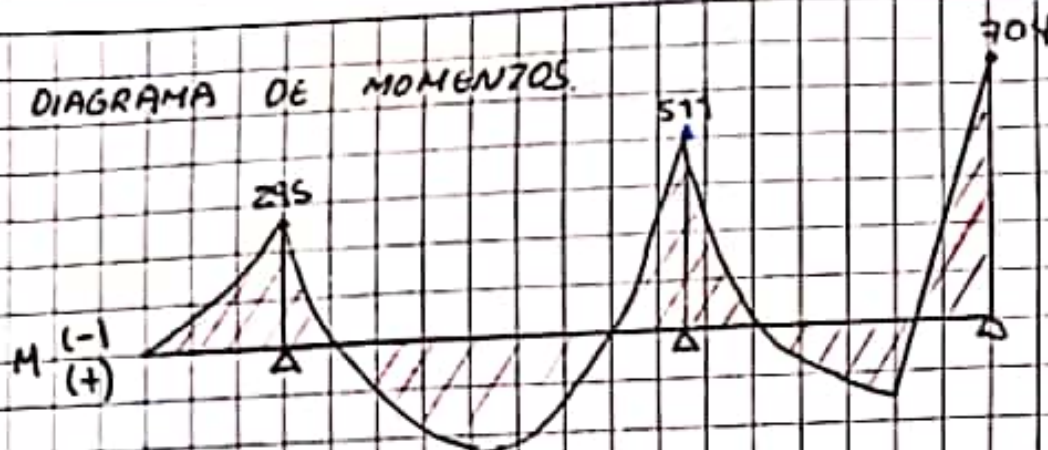
$$M_2(4) + 2M_3(4+0) = -7767,47$$

$$\begin{cases} 20(M_2) + 4(M_3) = -78093,75 \\ 4(M_2) + 8(M_3) = -7767,47 \end{cases}$$

$$M_2 = -511,83 \text{ (kg} \cdot \text{m)} ; M_3 = -714,27 \text{ (kg} \cdot \text{m)}$$

Escaneado con CamScanner

DIAGRAMA DE MOMENTOS.



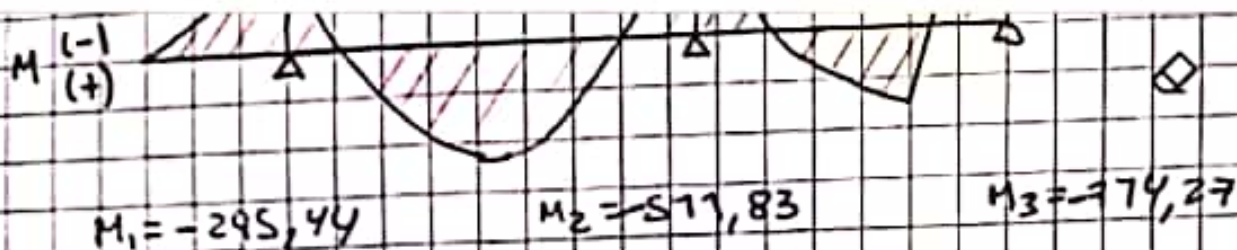
$$M_1 = -295,44$$

$$M_2 = -511,83$$

$$M_3 = -714,27$$

$$R_H = \frac{M_H - M_m}{L}$$

Escaneado con CamScanner



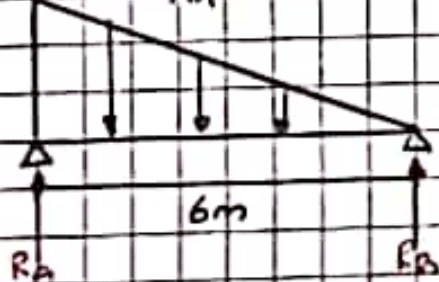
$$R_H = \frac{M_H - M_m}{L}$$

$$R_{H1} = \frac{511,83 - 295,44}{6} = 36,06$$

$$R_{H2} = \frac{774,27 - 511,83}{4} = 50,61$$

TRAMO AB

321,05 Ks/m



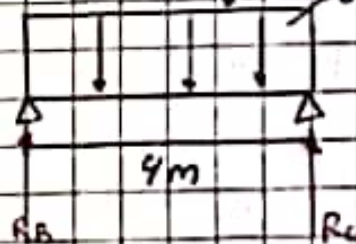
$$R_A = 642,10$$

$$R_B = 321,05$$

TRAMO BC

672 Ks

258 Ks/m



$$R_B = 693,08$$

$$R_C = 1004,02$$

$$R_T = R_{isos} \pm R_{hiper}$$

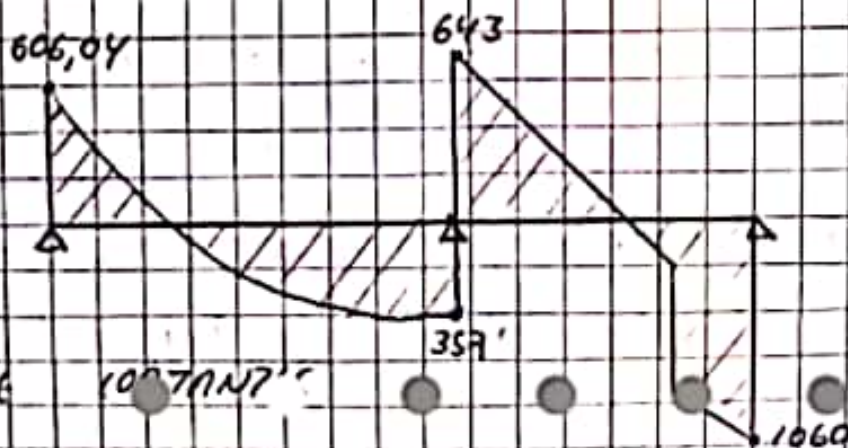
$$R_{T1} = 642,10 - 36,06 = 606,04$$

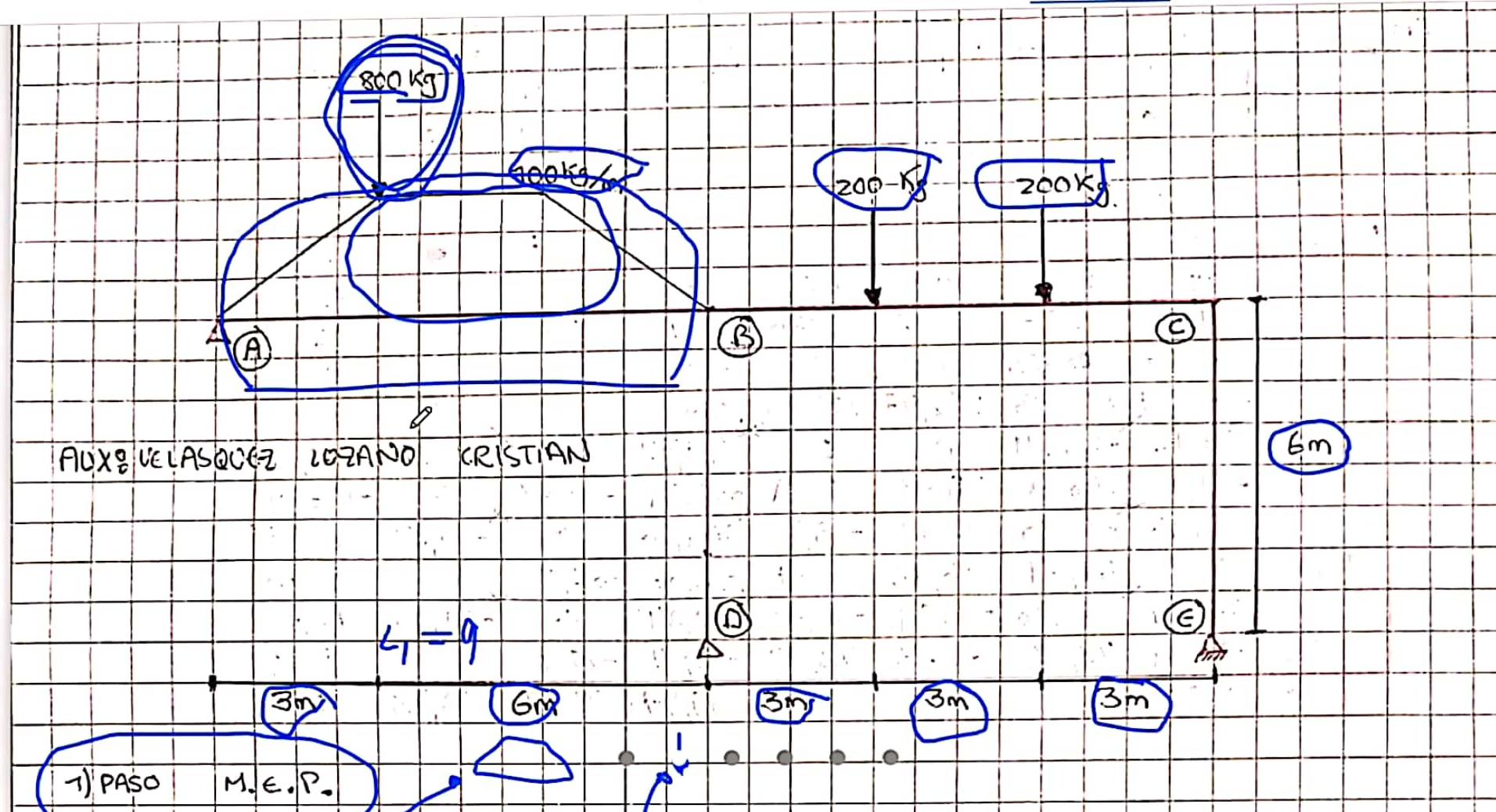
$$R_{T2} = 321,05 + 36,06 = 357,11$$

$$R_{T3} = 693,08 - 50,61 = 643,47$$

$$R_{T4} = 1004,02 + 50,61 = 1060,53$$

Q (+)  
(-)





AUX VELASQUEZ LOZANO CRISTIAN

6m

Diagrama de un sistema de vigas con dos tramos de longitud  $L = 9$  m. El primer tramo tiene una carga distribuida de  $100 \text{ N/m}$  y una carga puntual de  $200 \text{ N}$  a  $3 \text{ m}$  del apoyo B. El segundo tramo tiene una carga puntual de  $800 \text{ N}$  a  $3 \text{ m}$  del apoyo E. Los apoyos son un apoyo fijo en A y un apoyo móvil en E. Las distancias entre los apoyos y las cargas están indicadas en el diagrama.

1) PASO M.E.P.

$$M_{AB} = \frac{qL}{96} (L+b) \left( 5 - \frac{b^2}{L^2} \right) + \frac{Pa^2}{L^2} = \frac{100(9)}{96} (9+3) \left( 5 - \frac{3^2}{9^2} \right) + \frac{800(3)(6)^2}{9^2}$$

$$M_{AB} = 550 + 1066,67 = 1616,67$$

$$M_{BA} = 550 + 533,33 = 1083,33$$

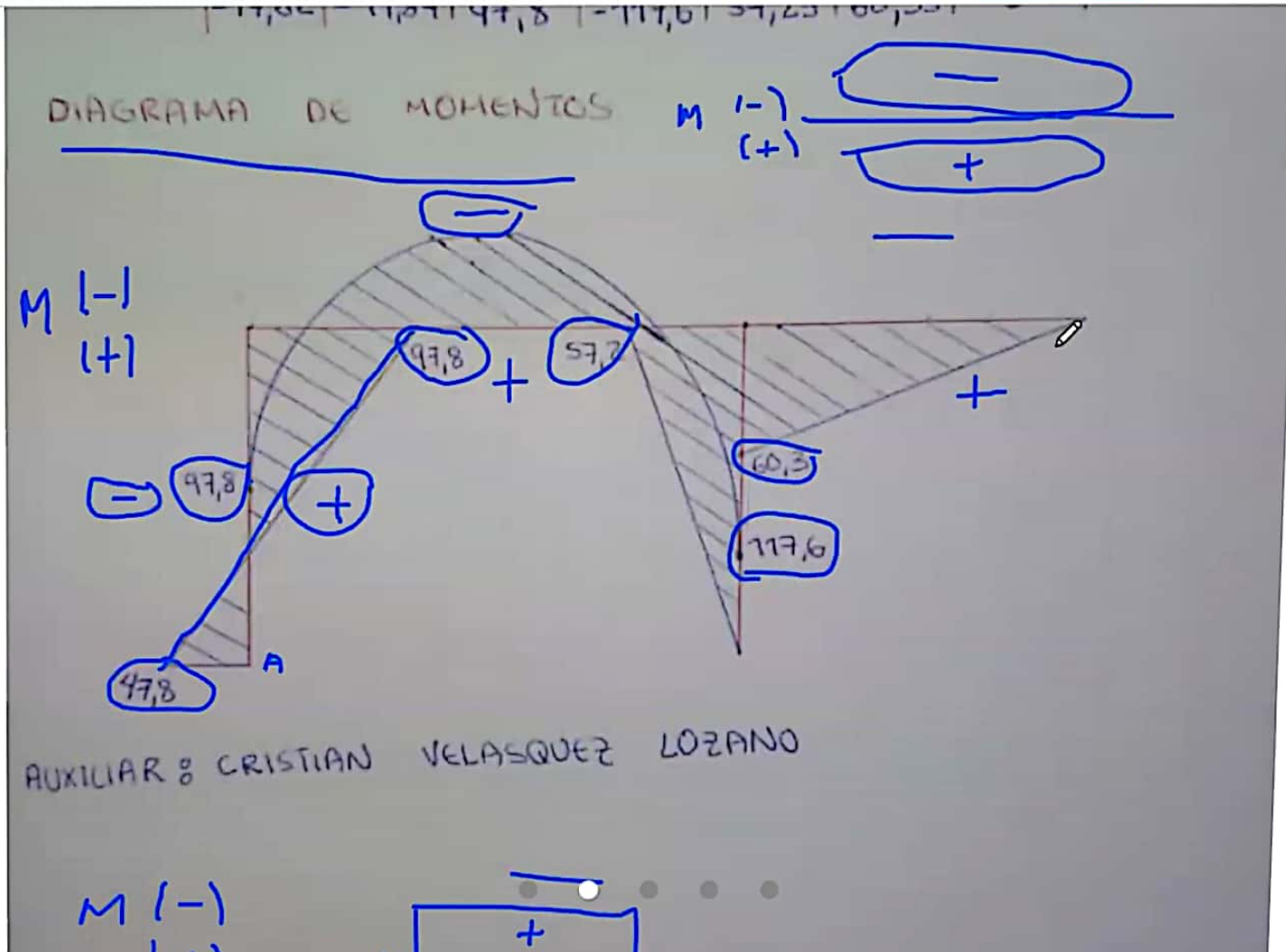
$$M_{DE} = M_{ED} = \frac{Pa(L-a)}{9} = \frac{200(9)(9-3)}{9} = 400$$

2) PASO F.D.

$$FD_{AB} = 1 \quad ; \quad FD_{BOB} = 1 \quad ; \quad FD_{EC} = 0$$

$$FD_{BA} = \frac{\frac{1}{9}}{\frac{1}{9} + \frac{1}{6} + \frac{1}{9}} = 0,286 \quad ; \quad FD_{BO} = \frac{\frac{1}{6}}{\frac{1}{9} + \frac{1}{6} + \frac{1}{9}} = 0,428$$

$$FD_{BC} = \frac{\frac{1}{9}}{\frac{1}{9} + \frac{1}{6} + \frac{1}{9}} = 0,286$$



$$FD_{CB} = \frac{\frac{1}{9}}{\frac{1}{9} + \frac{1}{6}} = 0,4$$

$$FD_{CE} = \frac{\frac{1}{6}}{\frac{1}{9} + \frac{1}{6}} = 0,6$$

| MODO   | A        | B        | C      | E      | D       |        |        |         |
|--------|----------|----------|--------|--------|---------|--------|--------|---------|
| TRAMO  | AB       | BA       | BC     | CB     | CE      | EC     | DB     |         |
| FD     | 1        | 0,286    | 0,428  | 0,286  | 0,4     | 0,6    | 0      | 1       |
| M.E.P. | 1616,67  | -1083,33 | 0      | 400    | -400    | 0      | 0      | 0       |
| 1ª IT  | -1616,67 | 195,43   | 292,47 | 195,43 | 160     | 240    | 0      | 0       |
|        | 97,72    | -808,34  | 0      | 80     | 97,72   | 0      | 120    | 146,24  |
| 2ª IT  | -97,72   | 208,31   | 311,73 | 208,31 | -39,09  | -58,63 | 0      | -146,24 |
|        | 104,16   | -48,86   | -73,72 | -19,55 | 104,16  | 0      | -29,32 | 155,87  |
| 3ª IT  | -104,16  | 40,48    | 60,57  | 40,48  | -41,66  | -62,5  | 0      | -155,87 |
|        | 20,24    | -52,08   | -77,94 | -20,83 | 20,24   | 0      | -31,25 | 30,29   |
| 4ª IT  | -20,24   | 43,12    | 64,53  | 43,12  | -8,096  | -12,14 | 0      | -30,29  |
|        | 0        | -1505,27 | 578,24 | 925,96 | -106,73 | 106,73 | 87,43  | 0       |